

Enhanced Reconstruction of Interacting Hands for Immersive Embodiment

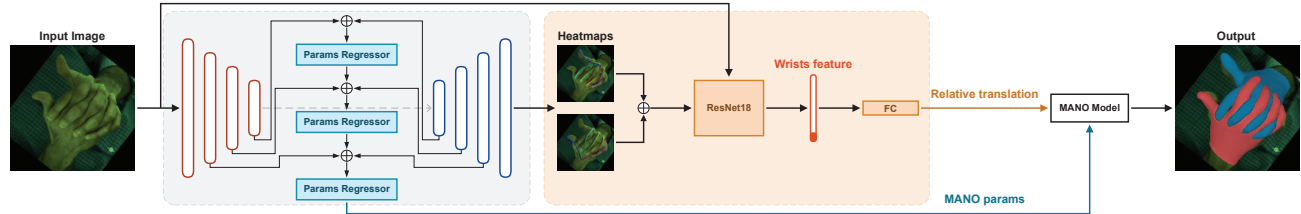
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Figure 1: Overview of the proposed method. Given an RGB image, we generate the two-hand relative translation and MANO parameters in parallel. The two-stream data is then fed into the MANO model to derive the ultimate recovered hand mesh.

ABSTRACT

Interacting hands reconstruction serves as a channel for natural user engagement in virtual reality, providing realistic embodiment that markedly elevates the immersive experiences. However, accurate prediction of the spatial relations between two hands remains challenging due to the severe occlusion and homogeneous appearance of hands. This paper presents a spatial relationship refinement method for hand reconstruction, employing a module to yield the 3D relative translation between hands and a novel loss function to limit hand mesh penetration. Our method achieves state-of-the-art performance on the InterHand2.6M dataset, offering considerable potential for interacting hands reconstruction in enhancing embodiment for virtual reality.

Index Terms: Computing methodologies—Artificial intelligence—Computer vision—Reconstruction

1 INTRODUCTION

Embodied interaction is one of the most pivotal research directions in the realm of augmented and virtual reality (AR/VR). Consequently, the reconstruction of interacting hands, a process that generates the 3D spatial models that accurately replicate an individual's actual dual hand pose and shape in 3D space, can effectively bolster the sense of embodiment [1, 4]. Despite notable progress made in interacting hands reconstruction, the task of interacting hands mesh recovery from a monocular RGB image is still fraught with challenges. Apart from the occlusion and structural resemblance between hands that impose great confusion upon the feature extractor, the deficiency of depth information from monocular images also potentially results in penetration of the hand mesh during the generating process, where the network is incapable of figuring out the precise relative position of the hands. Furthermore, in hand reconstruction tasks, the prevalent loss function, such as MANO loss, supervises the 2D/3D joint and mesh vertices coordinates, while the smooth loss maintains the fluidity of mesh. As these functions predominantly focus on the accuracy and consistency of predictions, they do not directly augment the spatial relation. To tackle the aforementioned challenges,

Zuo et al. [6] utilize the physics engine and a sampling-based optimization scheme for hand pose refinement, however, both of which may be computationally intensive and time-consuming. Regarding the design of the loss function, Ren et al. [5] use the offset loss to supervise the spatial relation of the hands and Lee et al. [2] introduce a penetration loss in the neural implicit representation, penalizing the overlapping two-hand occupancy values. Nonetheless, a loss function has not yet been incorporated into the parametric model to constrain the penetration issue.

Hence, in this paper, we propose an interacting hands reconstruction method that involves a relative position enhancement module and a boundary loss function to refine the spatial relation of interacting hands. As shown in Fig. 1, the proposed network comprises an image feature extractor for multi-scale feature extraction, a spatial relation enhancing module for refining the relative position of two hands, and a hand parameter regression module to recover MANO model parameters of each hand. Moreover, a novel boundary loss is proposed to improve the spatial relation, which harnesses a simply computed signed distance to constrain the penetration phenomenon. Evaluated on the InterHand2.6M dataset, our method achieves state-of-the-art performance and its efficacy in refining spatial relations holds significant potential for enhancing the sense of embodiment in virtual environments.

2 METHOD

Our framework comprises three components: a feature extraction module, a spatial relation enhancement module, and a hand parameter regression module.

2.1 Image Feature Extraction Module

Considering the great performance of HRNet in human pose estimation, we utilize it as the backbone for the encoder in this study. Given a monocular image $\mathbf{I} \in \mathbb{R}^{(3,256,256)}$, we yield a sequence of pyramid image features $\{\Phi_k, k = 1, 2, 3\}$ via HRNet. These features are then upsampled and concatenated together, from which we predict the heatmaps of joints $\mathbf{H}_{\text{left}}, \mathbf{H}_{\text{right}}$ through the decoder.

2.2 Spatial Relation Enhancement Module

Within this module, we effectuate the prediction of the spatial relative position of two hands by selectively extracting the features of both wrists, as the wrist serves as a crucial joint in representing spatial relationships among all hand joints. As demonstrated in the orange section of Fig. 1, we first ascertain the positions of the left

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and right wrists from the heatmaps, which undergo further processing with the raw image via a Resnet-18 network to extract relevant features. Bilinear interpolation is then employed to sample features at precise wrist coordinates. Subsequently, the extracted features are fed into a fully connected layer, enabling the prediction of relative position $t \in \mathbb{R}^3$ of the hands.

2.3 Hand Parameter Regression Module

The regressor generates MANO parameters, which include a shape parameter $\beta \in \mathbb{R}^{10}$ and a pose parameter $\theta \in \mathbb{R}^{16 \times 3}$ and are then forwarded to the MANO layer to generate the final 3D hand mesh. What's more, we integrate the graph convolutional network in the regression process to effectively capture the contextual information among keypoints, thereby achieving a more precise parameter regression.

2.4 Loss Functions

In our approach, we apply the following loss functions to supervise the network outputs. Initially, a novel boundary loss function is proposed to measure penetration extent, thereby optimizing the network. In addition to the boundary loss, this model also includes a reconstruction loss to supervise the generated hand joints and vertices, a MANO loss for the MANO parameters, and a mean square error (MSE) loss to supervise the 2D heatmap from the image feature extraction module.

Boundary Loss. This loss function aims to compute the approximate signed distance value of hand mesh to measure the penetration extent. We calculate the simplified signed distance value for both hands, where the signed distance is negative if the predicted position of the hand mesh infiltrates the interior of another hand model and conversely positive when there is no penetration between the two hand models. The construction process is outlined as follows.

For each vertex $x \in V_{left}$, we first define the corresponding signed distance to the right hand SD_{left} :

$$\text{distance}_{min} = \min_{y \in V_{right}} \|x - y\|_2 \quad (1)$$

$$\text{distance}_{center} = \|x - \bar{V}_{right}\|_2 \quad (2)$$

$$SD_{left} = \begin{cases} -\text{distance}_{min}, & \text{distance}_{center} < \text{distance}_{min} \\ \text{distance}_{min}, & \text{distance}_{center} \geq \text{distance}_{min} \end{cases} \quad (3)$$

where $\bar{V}_{right}$ is the mean coordinate of all vertices on the right hand, and the construction process for SD_{right} parallels that of SD_{left} .

Ultimately, the boundary loss is defined as Eq.4, where the *threshold* is set as 0.01 to constrain the vertices that show a signed distance value below 0.01, hence prompting the network to generate 3D hand shapes that minimize mutual penetration.

$$\mathcal{L}_{boundary} = \text{ReLU}(-SD_{left} - \text{threshold}) + \text{ReLU}(-SD_{right} - \text{threshold}) \quad (4)$$

Reconstruction Loss. We use L1 loss to supervise the 3D coordinates of hand vertices and joints, whereas the MSE loss is used to constrain the 2D projection of the vertices.

MANO Loss. We utilize the MSE loss to supervise the pose and shape parameters, which can limit the regressed shapes close to the ground truth shape and eliminate inappropriate joint twists.

The total loss is calculated as a weighted sum of the above individual losses, as shown in Eq.5, enabling balanced optimization of interacting hands recovery.

$$\mathcal{L}_{total} = \lambda_b \mathcal{L}_{boundary} + \lambda_h \mathcal{L}_{hmps} + \lambda_r \mathcal{L}_{recon} + \lambda_m \mathcal{L}_{mano} \quad (5)$$

where λ_b , λ_h , λ_r , λ_m are the loss weights, which are set to 100, 10, 10, and 0.1. Additionally, it's worth mentioning that λ_b should be moderately small to keep the resultant mesh orderly.

3 RESULT

We evaluate our method on the InterHand2.6M dataset with the state-of-the-art method IntagHand [3]. For quantitative evaluation, we adopt the mean per joint position error (MPJPE) and vertex error (MPVPE) to assess the 3D position of the joint and mesh vertex. Our method improves by 12.97% (7.65 mm vs. 8.79 mm) and 7.86% (8.32 mm vs. 9.03 mm) in MPJPE and MPVPE, respectively.

Furthermore, Fig. 2 provides a qualitative comparison from various perspectives, specifically chosen to highlight the penetration issues prevalent in the reconstructed hand mesh generated by IntagHand. Observations from these identical angles reveal that our proposed method markedly mitigates the extent of penetration.

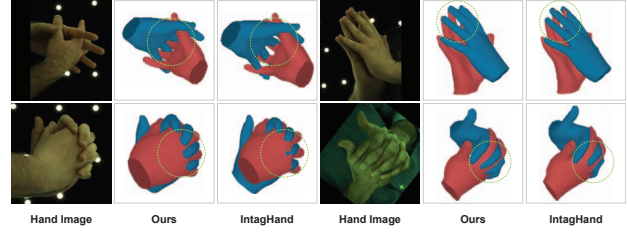


Figure 2: Visual comparisons with the state-of-the-art method IntagHand, our method has lower penetration extent.

4 CONCLUSION

We present a spatial relation refinement method for 3D interacting hands reconstruction. The enhancement module, which predicts relative translation, combined with a boundary loss, offers a computationally efficient solution for refining spatial relations. Integrating our novel boundary loss with an advanced dual-hand mesh recovery method can be a beneficial research direction for future interacting hands reconstruction from monocular images. This advancement would enhance the embodiment and realism within VR environments, thereby elevating the immersive experience.

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